

Question Bank on Data Analytics

1. What is Data Analysis? Explain characteristics of Data Analysis.
2. Explain the difference between Data Reporting and Data Analysis.
3. What are the applications of Data Analytics?
4. Define Data Analytics. Explain the process of Data Analysis.
5. What are the types of Data Analysis?
6. Explain the use of Python in Data Science.
7. Explain libraries used for Data Analysis.
8. What is NumPy? Explain array concept in NumPy.
9. Explain what is meant by "vectorized computation" in the context of array programming (e.g., using NumPy).
10. What is a NumPy ndarray? Describe its key attributes and explain how it differs from a regular Python list.
11. Explain what a multidimensional array object is in NumPy. How is it created, and how does NumPy internally represent and access its elements? Provide an example.
12. What are universal functions (ufuncs) in NumPy? List some common ufuncs and explain the concept of broadcasting in the context of ufuncs.
13. How does NumPy perform file I/O with arrays? Name functions used for saving/loading in binary and text formats, and list pros and cons of each.
14. What is pseudorandom number generation in NumPy? Which functions are used to generate random numbers, and how can reproducibility be ensured?
15. What are fast element-wise array functions in NumPy? How do they work, and why are they preferred over loops in array computations? Give examples.
16. What is NumPy? Why is it widely used in array-oriented programming?

By using Blooms Taxonomy

1. Demonstrate how the characteristics of Data Analysis are applied in business decision-making.
2. Explain the difference between Data Reporting and Data Analysis.
3. List the applications of Data Analytics in different fields.
4. Break down the Data Analysis process into stages and analyze their interdependence.
5. List the main types of Data Analysis.
6. Explain why Python is widely used in Data Science.
7. Define each library briefly used in Data Analysis.
8. Illustrate how to create one-dimensional and two-dimensional arrays in NumPy.
9. Explain how vectorized computation differs from using loops.
10. Define a NumPy ndarray. Explain the purpose of ndarray in NumPy.
11. Explain how NumPy represents a multidimensional array internally.
12. Describe the purpose of ufuncs in array-oriented programming.
13. Define file I/O in the context of NumPy arrays.
14. List NumPy functions used for random number generation.
15. Illustrate element-wise array functions in NumPy.
16. Describe the role of NumPy in array-oriented programming.